MIL-M-20344C (MU)
10 February 1969
SUPERSEDING
MIL-M-20344B (MU)
18 June 1965

MILITARY SPECIFICATION

MOTOR, ROCKET, M4
PARTS FOR

1. SCOPE

1.1 This specification covers the parts for one type of Rocket Motor designated as Motor, Rocket, M4.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-E-540 - Envelopes; Water Resistant, for Packing List and Shipping Documents.

MILITARY

MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing, and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1340

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DRAWINGS

ORDNANCE CORPS

7548183 - Box, Packing, Ammunition, for Rocket Motor, M4, Assembly Details and Packing
7548420 - Rocket Motor, M4, Metal Parts Assembly

PUBLICATIONS

ORDNANCE CORPS

ET-7548420 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Parts Assembly. - The parts assembly shall comply with all requirements specified on Drawing (dwg.) 7548420, and with all requirements specified in applicable specifications.

3.3 Pull Test. - The motor assembly unit shall withstand a 600 plus 50 pounds pull for a period of approximately 5 seconds without evidence of rupture or deformation when tested in accordance with 4.4.1.

3.4 Painting and Coating. - All interior and exterior coatings shall be applied with a continuous surface and shall comply with the requirements of applicable specifications.

3.5 Workmanship. - The item of shipment, its components, and materials shall be free of dirt, chips, grease, corrosions, and other foreign material. There shall be no cracks, scratches, burrs, dents, gouges, holes, porous areas, distortions, or other undesirable characteristics which would in any way, detract from the useability of the product for its intended purpose. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents. The painting and coating on all components shall be restored when such paintings and coatings have been marred or obliterated.

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3.6 First Article Inspection. - This specification makes provisions for First Article Inspection. Requirements for the submission of First Article Samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the preformance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Submission of Product. - At the time each completed lot ot items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. - The contractor shall submit a First Article Sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. The First Article Sample shall consist of five (5) complete assemblies and ten (10) complete sets of parts (i.e., 10 each of every individual component and every subassembly), which

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have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample including statements of findings. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A First Article Sample, or portions thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed. - Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection. - If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection Provisions.

4.3.1 Lot formation. - The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by

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one manufacturer, in one unchanged process using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specification, same specification revisions. Changes to the process, specification, or drawing, not affecting safety, performance, interchangeability, or storage, as determined by the Government shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination. - The examinations listed in the following subparagraphs shall be performed on inspection lots of components or assemblies designated in the applicable paragraph headings. Inspection for critical defects shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractors inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cover, Nozzle (see dwg. 7548436 covering a detail of dwg. 7548420).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 4.00 percent		
201.	Pitch diameter of threads, max...	Gage	01001
202.	Minor diameter of threads, max...	Gage	01002
203.	Diameter of counterbore, max.....	Gage	01003
204.	Depth of counterbore.....	Gage	01004
205.	Depth to end of thread, min.....	Gage	01005
206.	Diamond knurl missing.....	Visual	01006
207.	Phosphate coating missing or inadequate.....	Visual	01007
208.	Evidence of poor workmanship.....	Visual	01008

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4.3.2.2 Nozzle, Prior to Coating Interior (See dwg. 7548435 Covering a detail of dwg. 7548420)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Concentricity of expansion cone with pitch diameter and perpendicularity of expansion cone with shoulder	Gage	02001
102.	Diameter of throat	Gage	02002
103.	Pitch diameter of thread,max.	Gage	02003
104.	Minor diameter of thread,max.	Gage	02004
Minor:	AQL 4.00 percent		
201.	Pitch diameter, min.	Gage	02005
202.	Large inside diameter at exhaust end	Gage	02006
203.	Length of thread undercut max.	Gage	02007
204.	Depth of threaded cavity,min.	Gage	02008
205.	Total length, min.	Gage	02009
206.	Diameter of thread undercut, max.	Gage	02010
207.	Profile of entrance end of Venturi	Gage	02011
208.	Length of flat surface in throat excessive	Gage	02012
209.	Chamfer missing	Visual	02013
210.	Radius missing	Visual	02014
211.	Finish of throat improper	Visual	02015
212.	Phosphate coating missing or inadequate	Visual	02016
213.	Evidence of poor workmanship	Visual	02017

4.3.2.3 Nozzle, Igniter Prior to Coating Interior (See dwg. 7548438)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Concentricity of expansion cone with pitch diameter and perpendicularity of expansion cone to shoulder	Gage	03001

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102.	Diameter of throat	Gage	03002
103.	Pitch diameter of internal thread, max	Gage	03003
104.	Minor diameter of internal thread, max	Gage	03004
Minor:	AQL 4.00 percent		
201.	Diameter of sealing ring groove	Gage	03005
202.	Pitch diameter of external thread	Gage	03006
203.	Distance from rear exhaust end to thread	Gage	03007
204.	Width of sealing ring groove	Gage	03008
205.	Distance sealing ring groove to end, max	Gage	03009
206.	Length of thread undercut, max	Gage	03010
207.	Depth of threaded cavity, min	Gage	03011
208.	Large inside diameter at exhaust end	Gage	03012
209.	Total length, min	Gage	03013
210.	Diameter of thread undercut, max	Gage	03014
211.	Profile of entrance end	Gage	03015
212.	Length of flat surface in throat excessive	Gage	03016
213.	Radius missing	Visual	03017
214.	Chamfer missing	Visual	03018
215.	Finish of throat improper	Visual	03019
216.	Phosphate coating missing or inadequate	Visual	03020
217.	Evidence of poor workmanship	Visual	03021

4.3.2.4 Base, Trap (See dwg. 7548434)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.00 percent		
201.	Inside diameter, min	Gage	04001
202.	Outside diameter, max	Gage	04002

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4.3.2.5 Rod, trap (see dwg. 7548430)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Diameter of trap rod head, min	Gage	05001
Minor:	AQL 1.00 percent		
201.	Diameter head end to rod offset	Gage	05002
202.	Evidence of poor workmanship	Visual	05003

4.3.2.6 Trap, Weldment (see dwg. 7548429)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Braze or weld inadequate	Visual	06001
Minor:	AQL 1.50 percent		
201.	Concentricity of trap leg circle with outside diameter of trap base	Gage	06002
202.	Trap leg circle diameter, max	Gage	06003
203.	Evidence of poor workmanship	Visual	06004

4.3.2.7 Trap Assembly (see dwg. 7548428)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.50 percent		
201.	Total length, max	Gage	07001
202.	Powder retainer clip missing	Visual	07002
203.	Evidence of poor workmanship	Visual	07003

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4.3.2.8 Motor assembly prior to coating interior (see dwg. 7548421)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Pitch diameter of threads, min	Gage	08001
102.	Major diameter of threads, min	Gage	08002
103.	Perpendicularity of threads to block and location of threads with respect to block adapter hole	Gage	08003
104.	Parallelism of payload adapter block faces	Gage	08004
105.	Diameter of hole in payload adapter block, min	Gage	08005
106.	Weld on payload adapter block inadequate	Visual	08006
107.	Plug gage fails to enter motor tube completely	Gage	08007
Minor:	AQL 4.00 percent		
201.	Pitch diameter of threads, max	Gage	08008
202.	Major diameter of threads, max	Gage	08009
203.	Diameter of drilled hole in payload adapter block, min	Gage	08010
204.	Distance payload adapter block to end of tubes	Gage	08011
205.	Total length, min	Gage	08012
206.	Length of thread, min	Gage	08013
207.	Weld not flush with outside diameter of tube	Visual	08014
208.	Chamfer missing	Visual	08015
209.	Weld or spatter in payload block spot face area	Visual	08016
210.	Protective coating with bare spot	Visual	08017
211.	Evidence of poor workmanship	Visual	08018

4.3.2.9 Metal parts assembly prior to assembly of nozzle cover and coating (see dwg. 7548420).

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Categories	Defects	Method of Inspection	Code No.
Critical:			
	1. Hydrostatic test stamp missing from either nozzle or adapter block	Visual	09001
Major: None defined.			
Minor: None defined.			

4.3.2.10 Metal parts assembly (see dwg. 7548420)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Assembly dented or distorted	Visual	10001
102.	Nozzle cover missing	Visual	10002
Minor:	AQL 2.5 percent		
201.	Length end to nozzle, max	Gage	10003
202.	Total length, max	Gage	10004
203.	Bare spot in paint	Visual	10005
204.	Bare spot in interior coating	Visual	10006
205.	Oil in threads inadequate	Visual	10007
206.	Marking misleading or unidentifiable	Visual	10008
207.	Evidence of poor workmanship	Visual	10009

4.3.2.11 Sealed wood packing box (see dwg. 7548183)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Board broken or split	Visual	11001
102.	Hardward missing, torn or loose	Visual-Manual	11002
103.	Strapping missing, broken or loose	Visual-Manual	11003
Minor:	AQL 2.50 percent		
201.	Hardware improperly engaged	Visual-Manual	11004

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202.	Strapping improperly assembled	Visual-Manual	11005
203.	Contents loose	Manual	11006
204.	Screw loose or missing	Visual-Manual	11007
205.	Marking misleading or unidentifiable	Visual	11008

4.3.3 Testing

4.3.3.1 First Article.-The sample submitted as the first article quantity shall be tested as specified below after having been examined for all drawing dimensions and requirements. Rejection and re-submission shall be in accordance with 4.2.3.

4.3.3.1.1 Pull test - Motor assembly (see 3.3 and 4.4.1) - Code No. 08019.

4.3.3.1.2 Hydrostatic test (see dwg. 7548420 and 4.4.2) - Code No. 10010.

4.3.3.2 Production testing

4.3.3.2.1 Pull test - motor assembly (see 3.3) - Major defect - Code No. 08019.-This test shall be conducted 100 percent using the test equipment of 4.3.4 as specified in 4.4.1. Any motor assembly failing to meet the requirement of 3.3 shall be classed defective and removed from the lot.

4.3.3.2.2 Hydrostatic test (see dwg. 7548420) - Critical defect - Code No. 10010.-This test shall be performed 100 percent. The test shall be performed as specified in 4.4.2 using test equipment specified in 4.3.4. Any assembly which fails to meet the drawing requirements shall be classed defective and removed from the lot.

4.3.4 Inspection Equipment.-Equipment Tabulation Number ET-7548420 identifies the inspection equipment required to perform the examinations and tests perscribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

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4.4.1 Pull test.-The pull shall be applied to the payload adapter block in line with the longitudinal axis of the motor assembly and in a direction opposite to the flight direction.

4.4.2 Hydrostatic test.-The unpainted metal parts assembly with both trap weldments, shall be subjected to the internal hydrostatic test. (Discs of the same outside diameter and thickness at the base plate may be substituted for the weldments to obtain correct thread engagement.) The pressure shall be applied in such a manner that the assembly is free to expand in all directions and the full force is extended on the motor assembly threads. Water, containing a bright color agent to facilitate the visual detection of leaks, shall be used in the test. Assemblies shall be thoroughly dried immediately after the test.

5. Preparation for Delivery

5.1 Presentation and Packaging

5.1.1 Level C.-Four sealing rings shall be sealed in an envelope conforming with the requirements of Class 1, Style II of Specification PPP-E-540 except that the painting required by paragraph 3.6 shall be eliminated. The envelope shall be printed with the part number, quantity and nomenclature. One each envelope shall be packed in each wood packing box.

5.2 Packing

5.2.1 Level C.-Unless otherwise specified, the rocket motor metal parts shall be packed in accordance with dwg. 7548183.

5.3 Marking.-The box shall be marked in accordance with Standard MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.5).

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6.3 Inspection equipment. - The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL). - Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs

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shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs. - All submitted designs shall contain a reference to the applicable five-digit Code Number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND14. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

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